

UpSkillr : Online Learning and Skill Development Platform

Software Requirements Specification

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Prepared By	Product Engineering
Reviewed By	Technical Lead, Delivery

Revision History

Version	Summary of Change	Author
1.0	Initial draft circulated for internal review	Product Engineering
1.1	Scope refined following technical review	Technical Lead
1.2	Functional requirement identifiers, non-functional requirements and acceptance criteria added	Product Engineering

1. Overview

UpSkillr is a full-stack web platform that simplifies online learning and personal skill growth. It allows instructors to create short courses and learning modules while enabling learners to explore, enrol, and track their progress.

The platform is a lightweight Learning Management System built on the MERN stack, focusing on functionality, scalability, and clean interface design.

2. Project Goals

- Build a complete MERN-based learning management platform supporting two user roles: Instructor and Learner.
- Implement course creation, enrolment, and progress tracking features.
- Allow instructors to upload lessons, resources, and assessments.
- Provide learners with progress visualisation and course completion status.

- Design a modern, responsive, and intuitive user interface.

3. Technical Scope

The solution shall be developed using **MongoDB, Express, React, and Node.js**. Authentication and role-based access control shall be implemented using **JWT**. All major operations, including course management, user profile management and progress updates, shall be handled through RESTful APIs.

4. Functional Requirements

Priorities follow the MoSCoW convention. **Must** requirements are mandatory for acceptance; **Should** requirements are expected unless justified; **Could** requirements are desirable enhancements.

ID	Requirement	Priority
FR-01	The system shall provide secure login and registration for instructors and learners.	Must
FR-02	The system shall enforce role-based access appropriate to each user type.	Must
FR-03	The system shall allow instructors to create, edit and publish courses.	Must
FR-04	The system shall allow instructors to upload lessons, resources and assessments to a course.	Must
FR-05	The system shall allow learners to browse available courses.	Must
FR-06	The system shall allow a learner to enrol in a course through a single action.	Must
FR-07	The system shall track lesson completion and overall course progress for each learner.	Must
FR-08	The system shall present progress and course completion status on the learner dashboard.	Must
FR-09	The system shall allow learners to rate completed courses and submit feedback.	Should
FR-10	The system shall provide an administrative module to review courses and monitor system activity.	Could
FR-11	The system shall present a responsive interface supporting both roles.	Must

5. Non-Functional Requirements

ID	Category	Requirement
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NFR-01	Performance	The application shall respond to user actions without a delay noticeable during normal use.
NFR-02	Scalability	The application shall remain responsive as the number of records and concurrent users grows.
NFR-03	Security	All API endpoints shall enforce authentication and role-based authorisation. Credentials, tokens and connection strings shall not be committed to source control.
NFR-04	Reliability	Data submitted by users shall be persisted such that work is not lost in the event of an unexpected disconnection.
NFR-05	Usability	The interface shall be operable without prior training and shall provide visible feedback for every user action.
NFR-06	Compatibility	The application shall function on current versions of major desktop browsers and shall remain usable at tablet screen sizes.
NFR-07	Accessibility	Interactive controls shall be keyboard reachable and clearly labelled.
NFR-08	Maintainability	Code shall follow a consistent style with modular components, and environment setup shall be documented in the repository.

6. Acceptance Criteria

The following criteria shall be demonstrated during user acceptance testing.

ID	Ref	Criterion
AC-01	FR-02	Given a user logged in as a learner, when they attempt to open course creation, then access is refused.
AC-02	FR-03	Given an instructor, when they publish a course, then the course becomes visible to learners browsing available courses.
AC-03	FR-06	Given a published course, when a learner enrolls, then the enrolment is recorded and the course appears on their dashboard.
AC-04	FR-07	Given an enrolled learner, when they mark a lesson complete, then the recorded progress for that course is updated.
AC-05	FR-08	Given a learner who has completed all lessons in a course, when they open their dashboard, then the course is shown as complete.

AC-06	FR-09	Given a completed course, when the learner submits a rating and feedback, then it is stored against that course.
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7. Mid-Point Review

A mid-point review shall be conducted at approximately 60% of the delivery period. Each team shall submit the following:

- **Repository link** — public, or with access granted to the review team, containing the complete commit history for the project to date.
- **Demo video** — a screen recording of three to five minutes showing the running application. Slides and design mockups shall not be accepted in place of a recording.
- **Status note** — a short summary of completed requirements, work in progress, and any blockers or scope concerns.

The submission shall be assessed against the Must-priority requirements listed in Section 4. The following shall be demonstrable at the mid-point review: **FR-01, FR-03, FR-05, FR-06**.

Commit history shall be reviewed for evidence of contribution distributed across the team and of sustained work across the delivery period.

Requirements not evidenced at the review shall be recorded and re-planned across the remaining delivery period. Review outcomes shall be communicated to the team leader.

8. Expected Deliverables

- A working prototype of the UpSkillr platform with complete learning and enrolment functionality.
- Backend APIs supporting authentication, course CRUD operations, and enrolment logic.
- A fully responsive React frontend supporting both roles with a simple and modern layout.
- Technical documentation outlining architecture, data models, and API specifications.
- Presentation material summarising the platform features and technical implementation.

9. Expected Outcomes

- A functional and visually appealing learning platform suitable for real-world deployment.
- Consolidated understanding of MERN stack development, RESTful API integration, and authentication workflows.
- Established practice in modular coding, efficient state management, and database optimisation.
- A demonstration of team collaboration and agile workflow management.
- A project reflecting modern web application standards used in the edtech industry.

10. Approach Guidelines

- Start with user flow mapping for both the Instructor and Learner modules.
- Design database collections for Users, Courses, Lessons, and Enrolments before backend implementation.
- Prioritise authentication and core CRUD features before interface enhancements.
- Conduct testing on enrolment, progress tracking, and access permissions.
- Prepare a short demonstration showcasing both instructor and learner perspectives.